2nd Order DEs: Simple Harmonic Motion

Questions 1–39

Question 1. It is given that $x = 3\cos 2t - 4\sin 2t$.

a) Find $\dot{x}$.

b) Find $\ddot{x}$.

c) Show that $\ddot{x} = -\omega^2 x$, stating the value of ω.

Question 2. It is given that $x = A\cos 3t + B\sin 3t$.

a) Show that $\ddot{x} = -\omega^2 x$, stating the value of ω.

b) Given that $x = 2$ and $\dot{x} = -3$ when $t = 0$, find the values of A and B.

Question 3. Show that the general solution to the differential equation $\ddot{x} + \omega^2 x = 0$, where ω is a positive constant, is $x = A\cos\omega t + B\sin\omega t$.

Question 4. Solve the differential equation $\ddot{x} + \omega^2 x = 0$ given that $x = x_0$ and $\dot{x} = v_0$ when $t = 0$.

Question 5. Solve the following differential equations, subject to the given initial and boundary conditions.

a) Given that $\ddot{x} = -36x$, where $x = 3$ when $t = 0$ and $x = 4$ when $t = \frac{\pi}{12}$.

b) Given that $\ddot{x} = -400x$, where $x = -5$ when $t = 0$ and $x = 8$ when $t = \frac{3\pi}{40}$.

c) Given that $\ddot{x} = -2x$, where $x = 1$ when $t = 0$ and $\dot{x} = \frac{1}{3}$ when $t = 0$.

d) Given that $25\ddot{x} = -x$, where $x = \frac{1}{20}$ when $t = 0$ and $\dot{x} = \frac{1}{100}$ when $t = 10\pi$.

Question 6. Sketch the following graphs for $t \geq 0$.

a) $x = 3\cos t + 4\sin t$

b) $x = 4\cos 2t - 5\sin 2t$

c) $x = 4\sin 3t - 2\cos 3t$

Question 7. It is given that $\ddot{x} = -9x$, where $x = 2$ when $t = 0$ and $\dot{x} = 6$ when $t = \frac{\pi}{3}$.

a) Solve the differential equation.

b) Sketch a graph of x against t.

Question 8. It is given that $x(t) = A\cos\omega t + B\sin\omega t$.

a) Show that $x\left(t + \frac{2\pi}{\omega}\right) = x(t)$.

b) State the period of $x(t)$.

Question 9. It is given that $A\cos\omega t + B\sin\omega t \equiv R\cos(\omega t - \phi)$, where $R \geq 0$ and $0 < \phi < \frac{\pi}{2}$. Find the values of R and ϕ in terms of A and B.

Question 10. It is given that $\ddot{x} = -\omega^2 x$.

a) Explain what this says about the relationship between acceleration and displacement.

b) i) If $x > 0$, what is the sign of $\ddot{x}$?

ii) If $x < 0$, what is the sign of $\ddot{x}$?

iii) What does this imply about the motion relative to $x = 0$?

Question 11. A particle satisfies $\ddot{x} = -7x$.

a) State the relationship between acceleration and displacement.

b) Explain why $x = 0$ is an equilibrium position.

c) Describe the direction of motion immediately after passing through $x = 0$ with positive velocity.

Question 12. A particle satisfies $\ddot{x} = -\omega^2 x$. At some instant, $x > 0$ and $\dot{x} < 0$.

a) What is the sign of the acceleration?

b) Is the particle moving towards or away from equilibrium?

c) Explain whether the speed is increasing or decreasing.

Question 13. It is given that $x(t) = R\cos(\omega t - \phi)$.

a) Find $\dot{x}(t)$.

b) State the maximum speed.

c) Determine when turning points occur, in terms of $\omega t - \phi$, and the value of x at these points.

Question 14. A particle moves so that $x = R\cos(\omega t - \phi)$.

a) Show that when $t = \frac{\phi}{\omega}$, the particle is at maximum displacement.

b) Show that when $t = \frac{\phi}{\omega} + \frac{\pi}{2\omega}$, the particle passes through equilibrium.

c) Explain what the constant ϕ represents physically.

Question 15. A particle moves so that $x = R\cos\omega t$.

a) Find the velocity v of the particle in terms of t.

b) Using calculus, show that the maximum speed is ωR.

c) Explain why the maximum speed occurs at equilibrium.

Question 16. A particle moves along a line with displacement x and velocity $\dot{x}$ at time t.

a) Show that $\ddot{x} = \frac{dv}{dt}$.

b) Deduce that $\ddot{x} = v\frac{dv}{dx}$.

Question 17. A particle moves so that $\ddot{x} = -\omega^2 x$. By writing $\ddot{x}$ as $v\frac{dv}{dx}$, show that $v^2 + \omega^2 x^2 = c$ for some constant c.

Question 18. A particle moves so that $v^2 + \omega^2 x^2 = c$, where v is the velocity of the particle and x is the displacement over time. It is given that the amplitude of the motion is R.

a) Explain why $v = 0$ when $|x| = R$.

b) Use $x = \pm R$ and $v = 0$ to show that $v^2 + \omega^2 x^2 = \omega^2 R^2$.

Question 19. A particle moves so that $v^2 + 9x^2 = 36$.

a) Show that the motion is simple harmonic.
b) Find the angular frequency.
c) Find the amplitude.
d) Find the maximum speed.

Question 20. A particle moves so that $\ddot{x} = -25x$. Initially the particle is 0.2 m from equilibrium and is instantaneously at rest.

a) Find x as a function of t.
b) Find the period of x.
c) Find the maximum speed.
d) Find the first time the particle passes through equilibrium.

Question 21. A particle moves along a straight line and satisfies $\ddot{x} = -36x$. The greatest displacement from the centre is 0.5 m.

a) Find the period of the motion.

b) Find the greatest speed of the particle.

Question 22. A particle moves with SHM and satisfies $\ddot{x} = -4x$. Its maximum speed is 8 m s^{-1}.

a) Find the period of the motion.

b) Find the amplitude of the motion.

Question 23. A particle moves with SHM and satisfies $100\ddot{x} = -x$. When the particle is 0.3 m from the centre, its speed is 0.04 m s^{-1}.

a) Find the angular frequency of the motion.
b) Find the amplitude of the motion.
c) Find the period of the motion.
d) Find the speed when $x = 0.4$ m.

Question 24. A particle starts from the centre of oscillation with speed 5 m s^{-1}. It performs SHM with period $\frac{\pi}{2}$ seconds.

a) Find the angular frequency of the motion.
b) Find the amplitude of the motion.
c) Find the first time the particle reaches a displacement of 1 m.

Question 25. A particle moves so that $\ddot{x} = -16x + 32$, where displacement x is measured in metres, and time t is measured in seconds. At time $t = 0$, it is given that $x = 3$, and the particle is instantaneously at rest.

a) Find the equilibrium position.
b) Find the amplitude of the motion.
c) Find the period of the motion.

Question 26. A particle moves so that $\ddot{x} = -9x + 27$. At time $t = 0$, $x = 5$ and the particle is moving with speed 3 m s^{-1} towards the equilibrium position.

a) Find the equilibrium position.
b) Show that the motion is simple harmonic and find the angular frequency.
c) Find the amplitude of the motion.
d) Find the maximum speed of the particle.

Question 27. An offshore weather buoy is floating on the surface of the sea. It oscillates vertically in simple harmonic motion about its equilibrium level O. The motion has a period of 8π seconds and amplitude 1.5 metres. Initially the buoy is at O. Immediately after passing upward through O, the buoy rises to a point A which is 0.5 metres above O, and later reaches a point B which is 1.25 metres above O. Find the time taken for the buoy to travel from A to B.

Question 28. A camera mounted on a stabilising arm moves horizontally back and forth in simple harmonic motion about its central position C. When the camera is 0.4 metres from C, its speed is 1.2 m s^{-1}. When it is 0.8 metres from C, its speed is 0.6 m s^{-1}.

a) Find the amplitude of the motion.

b) Find the period of the motion.

Question 29. The end of a vibrating metal ruler moves with simple harmonic motion about a fixed point. When the tip is 15 mm from its central position, its speed is 120 mm s^{-1}. When it is 25 mm from the centre, its speed is 80 mm s^{-1}.

a) Find the amplitude of the vibration.

b) Find the period of the vibration.

Question 30. A cart of mass 2 kg moves along a smooth straight track. When it is x metres from a fixed point O, it experiences a force towards O of magnitude $18x$ N.

a) Form a differential equation for x in terms of t.

b) Show that the motion is SHM and find ω.

c) Find the period of the motion.

Question 31. A small trolley of mass 0.5 kg moves on a smooth horizontal surface. A control unit applies a force towards the point O of magnitude $2x$ N when the trolley is x metres from O. The trolley is pulled to $x = 0.3$ m and released from rest at $t = 0$ seconds.

a) Show that $\ddot{x} = -4x$.
b) Find x in terms of t.
c) Find the first time the trolley reaches O.
d) Find the speed of the trolley when $x = 0.2$ m.

Question 32. A light elastic spring of natural length 50 cm is fixed to a point O on a smooth horizontal table. A small block of mass 0.8 kg is attached to the other end. The block is pulled 0.25 m from its equilibrium position and released from rest. When the block is displaced x metres from equilibrium, the tension in the spring has magnitude $12x$ N and acts towards equilibrium.

a) Show that the motion is simple harmonic.
b) Find the period of the motion.
c) Find the maximum speed of the block.

Question 33. A block of mass 0.5 kg is attached to a light elastic spring on a smooth horizontal table. When the block is displaced x metres from its equilibrium position, the spring exerts a restoring force of magnitude $8x$ N. The block passes through equilibrium with speed 1.2 m s^{-1}.

a) Show that the motion is SHM and find ω.
b) Find the amplitude of the motion.
c) Find the maximum extension of the spring.

Question 34. A box of mass 2 kg slides on a smooth horizontal surface attached to a light elastic spring. The displacement x metres from equilibrium satisfies $\ddot{x} = -9x$. At time $t = 0$, the box is 0.5 metres from equilibrium and moving towards equilibrium with speed 1.5 m s^{-1}.

a) Find an expression for x in terms of t.
b) Find the amplitude of the motion.
c) Find the maximum speed of the box.

Question 35. A parcel of mass 1.2 kg rests on a smooth horizontal table and is attached to a light elastic spring fixed at one end. When the parcel is displaced x metres from equilibrium, the spring exerts a restoring force of magnitude $6x$ N. The parcel is pulled 0.6 m from equilibrium and released from rest.

a) Form a differential equation to model the motion of the parcel.
b) Solve the differential equation formed in (a) to find x in terms of t.
c) Find the time the parcel first returns to equilibrium.
d) State three modelling assumptions you have used to answer this question.

Question 36. A parcel of mass 1 kg rests at the midpoint O of two identical light elastic springs on a smooth horizontal surface. The other ends of the springs are attached to fixed points A and B, which lie on a straight line through O and are 3 metres apart. If the parcel is displaced x metres from O, each spring exerts a restoring force of magnitude $5x$ N towards O. The parcel is pulled 0.4 m towards B and released from rest.

a) Show that the motion is SHM.
b) Find the period of the motion.
c) Find the speed when the parcel is 0.2 metres from O.

Question 37. A small package of mass 2 kg lies on a smooth horizontal table at a midpoint O of two identical light elastic springs. The other ends of the springs are fixed at points A and B on a straight line, with $AO = BO = 1.5$ m. When the package is displaced x metres from O towards B, each spring exerts a force of magnitude $4x$ N on the package directed towards O. The package is pulled 0.6 m towards B and released from rest at $t = 0$.

a) Show that the package performs SHM and find ω.
b) Find the period of the motion.
c) Find the speed of the package when it is 1.1 metres from A.

Question 38. A small metal puck of mass m kg rests at a point O on a smooth horizontal air table. The puck is attached to two identical light springs along a straight line through O. The fixed ends of the springs are at points A and B, which are 2 m apart, with O the midpoint of AB. The puck is given a sharp tap so that it starts moving from O towards A with speed 0.8 m s^{-1}. At time t seconds, its displacement from O is x metres (take x to be positive in the direction from O towards A). Each spring exerts a force of magnitude $9mx$ N on the puck, directed towards O. The puck is modelled as a particle. The table is smooth and the only horizontal forces on the puck are due to the springs. Find the minimum distance from the puck to A during the subsequent motion.

Question 39. A small trolley of mass m kg moves on a smooth horizontal track. It is attached to two light elastic springs along a straight line. The left-hand spring is fixed to a wall at A, and the right-hand spring is fixed to a wall at B, where $AB = 2.4$ m. When the trolley is at a distance x metres from A, the left-hand spring exerts a force of magnitude $10mx$ N towards A, and the right-hand spring exerts a force of magnitude $5m(2.4 - x)$ N towards B. The trolley is released from rest at a position 0.3 m to the right of its equilibrium position.

a) Find the equilibrium position.

b) Show that the subsequent motion is simple harmonic and find ω.

c) Find, in terms of m, the greatest force exerted by the right-hand spring during the motion.